Tsung-Min (Vincent) Pai

tsungmip@andrew.cmu.edu | +1 (412) 224-3420 | www.linkedin.com/in/tsung-ming-pai | github.com/Bai1026

EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering — *Dec 2027*

- **Current Courses:** Foundations of Computer Systems, Systems & Toolchain for AI Engineers, Reliable Distributed Systems

National Taiwan University — Taipei, Taiwan
Bachelor of Science in Engineering, Electrical Engineering — *Dec 2025*

- **Relevant Courses:** Machine Learning, Generative Artificial Intelligence, Data Visualization, Data Structure, Algorithms

WORK EXPERIENCE

LOGOLO International Inc. — New Westminster, Canada
AI Engineer | Contractor — *Sep 2025 - Aug 2026*

- Built an AI corrector for **national legal exams** on rubrics co-defined with practicing lawyers, delivering real-time feedback at 95% expert-agreement accuracy and cutting grading turnaround 90% via a parallelized multi-agent pipeline.
- Shipped the system as a vendor-ready product, **packaging 8 subject-specific** FastAPI services behind a unified gateway into an encrypted Docker image with a versioned public API spec, enabling third-party integration without source access.

Genibuilder Inc. — Taipei, Taiwan
Founding AI Engineer | Full Time — *Sep 2024 - Aug 2025*

- Engineered a **conversational AI agent** with sales skills to replace aesthetic clinic consultants, automating the advisory process for real-time response and reducing client operational costs by 80%.
- Built an evaluation system **simulating user-product interactions**, reducing manual validation time by over 80%.
- Developed an AI-powered **data extraction** pipeline to automatically deliver authentic client chat logs for the R&D team.

Compal Electronics — Taipei, Taiwan
AI Project Engineer | Intern — *Jul 2024 - Aug 2024*

- Conducted research on knowledge graphs, GraphRAG, LLM agents, and fine-tuning for product improvement.

PUBLICATIONS & PROJECTS

BILLY: Steering Large Language Models via Merging Persona Vectors for Creative Generation
National Taiwan University | Advisor: Prof. Hung-Yi Lee, Prof. Shao-Hua Sun | ***EACL 2026 Oral****, First Author*

- Replaced multi-agent discussion with **activation layer vector merging** inside a single small LLM, raising originality 15% while cutting inference cost and time by 95% versus multi-agent system baseline.

LLM Discussion: Enhancing the Creativity of Large Language Models via Discussion Framework and Role-Play
National Taiwan University | Advisor: Prof. Hung-Yi Lee, Prof. Shao-Hua Sun | ***COLM 2024***

- Implemented a multi-agent role-play **discussion framework**, raising LLM creativity scores 20% over prior methods.
- Devised an LLM creativity **benchmark** reaching 0.7 correlation with human raters over 1,400+ generations.

EltAI Engine — Multimodal RAG Sports Agent
GenAI Stars 2024 Hackathon | Team Leader | High Distinction Award, Corporate Award

- Led a 4-person team building a multimodal RAG agent for live game analysis, multilingual Q&A, and outcome prediction.

LEADERSHIP & ACTIVITIES

Introduction to Generative AI, National Taiwan University — Taipei, Taiwan
Teaching Assistant / Worked with 30+ TAs for 1,000+ students / Contributed to an EMNLP paper — *Sep 2024 - Aug 2025*

Department Basketball Team, NTUEE — Taipei, Taiwan
Coach / Vice Captain — *Sep 2021 - Aug 2025*

SKILLS

Programming Languages: Python, C/C++, Go, MATLAB, JavaScript, SQL

LLM & Agents: LangChain, LlamaIndex, DSPy, AutoGen, Ollama, Hugging Face, RAG/GraphRAG

Machine Learning & Data: PyTorch, scikit-learn, NumPy, Pandas, Weights & Biases

Databases & Infrastructure: PostgreSQL, Neo4j, MongoDB, Redis, Chroma, Docker, GCP, FastAPI, CI/CD, Git, pytest